TEACHER COPY - MARK SCHEME

P1 exam: Days 1-6

60 marks with method guidance and diagnostic bands

YOUR AIM TODAY

Mark consistently, identify the first wrong step, and turn the result into the next study priorities.

YOUR RULE

Do not give this document to Rohan before the exam is complete. Award method marks when a correct method is clearly shown, even after a numerical slip.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

MARKING PRINCIPLES

How to use this scheme

M=method, A=accuracy, B=independent statement.

- Award an accuracy mark only when the required preceding method is present, unless the answer is clearly obtained correctly.
- Accept algebraically equivalent exact forms.
- Do not penalise the same arithmetic slip repeatedly; follow through where the method remains valid.
- Record each error as concept, method, algebra, sign, notation or checking.

GRADE GUIDE FOR THIS CHECKPOINT

51-60: secure A-level trajectory for these topics. 42-50: strong but repair specific gaps. 33-41: developing; reteach two weakest topics. Below 33: rebuild foundations before advancing.

Topic	Marks	Rohan's score
Quadratics	24	
Functions	10	
Coordinate geometry	12	
Circles	14	
Total	60	

QUESTIONS 1-3

Quadratics mark scheme

Total: 24 marks

Part	Answer and allocation	Marks
1(a)	2x^2+5x-12. M1 correct expansion; A1 simplified.	2
1(b)	(x-4)^2-5. M1 forms (x-4)^2; A1 constant -5.	2
1(c)	(4,-5). B1.	1
1(d)	x=4+-sqrt(5). M1 uses completed square or formula; A1 square-root step; A1 both exact roots.	3
2(a)	(3x+2)(x-3)=0, so x=-2/3 or 3. M1 factorises/uses formula; A1 each root.	3
2(b)	b^2-4ac; zero means one repeated real root / tangent to x-axis. B1+B1.	2
2(c)	36-4k=0, so k=9. M1 sets discriminant to zero; A1 substitutes; A1 solves.	3
3(a)	(x-4)(x+3)<0, so -3<x<4. M1 roots; A1 correct region; A1 strict notation.	3
3(b)	(2x-1)(x+3)>=0, so x<=-3 or x>=1/2. M1 roots; A1 regions; A1 endpoints included.	3
3(c)	p^2-36<0, so -6<p<6. M1 discriminant condition; A1 solution.	2

DIAGNOSTIC

If Q1 is weak: completing square/algebra. If Q2 is weak: discriminant. If Q3 is weak: roots plus parabola sign regions.

QUESTION 4

Functions mark scheme

Total: 10 marks

Part	Answer and allocation	Marks
4(a)	$f(-3)=-11$. B1.	1
4(b)	$fg(x)=f(g(x))=2(x^2+1)-5=2x^2-3$. M1 correct order; A1.	2
4(c)	$gf(x)=g(f(x))=(2x-5)^2+1=4x^2-20x+26$. M1 substitution; A1 expansion.	2
4(d)	$2x^2-3=15$, $x^2=9$, so $x=\pm 3$. M1 equation; A1 both roots.	2
4(e)	$f^{-1}(x)=(x+5)/2$. M1 reverses/rearranges; A1.	2
4(f)	g is not one-to-one: $g(a)=g(-a)$, for example $g(1)=g(-1)$. B1.	1

DIAGNOSTIC

Wrong fg/gf order needs composite-machine practice. A wrong inverse needs swap-and-rearrange practice. Part (f) tests conceptual understanding, not calculation.

QUESTION 5

Coordinate geometry mark scheme

Total: 12 marks

Part	Answer and allocation	Marks
5(a)	$m=(7-(-1))/(8-2)=8/6=4/3$. M1 formula; A1.	2
5(b)	$((2+8)/2,(-1+7)/2)=(5,3)$. B1.	1
5(c)	$\sqrt{6^2+8^2}=10$. M1 distance method; A1.	2
5(d)	$y+1=(4/3)(x-2)$, hence $y=(4/3)x-11/3$. M1 point-gradient; A1 substitution; A1 final form.	3
5(e)	Perpendicular gradient$=-3/4$ and midpoint$=(5,3)$. $y-3=(-3/4)(x-5)$, e.g. $3x+4y=27$. M1 negative reciprocal; M1 midpoint used; A1 equation; A1 simplified equivalent.	4

DIAGNOSTIC

Separate formula recall from sign errors. The perpendicular bisector requires both ideas: negative reciprocal gradient and midpoint.

QUESTION 6

Circles mark scheme

Total: 14 marks

Part	Answer and allocation	Marks
6(a)	$(x-3)^2+(y+2)^2=25$. M1 completes x square; M1 completes y square; A1 equation.	3
6(b)	Centre (3,-2), radius 5. B1+B1.	2
6(c)	At (6,2): $(6-3)^2+(2+2)^2=9+16=25$, so P lies on C. M1 substitution; A1 conclusion.	2
6(d)	Radius gradient=(2-(-2))/(6-3)=4/3. M1+A1.	2
6(e)	Tangent gradient=-3/4; y-2=(-3/4)(x-6), e.g. 3x+4y=26. M1 gradient; M1 point-gradient; A1.	3
6(f)	Set y=-2: $(x-3)^2=25$, so x=8 or -2. Points (8,-2) and (-2,-2). M1 equation; A1 both points.	2

Correction prescription

Score on topic	Required response
80% or more	One challenge question; continue.
60-79%	Redo every lost mark, then complete three targeted questions.
Below 60%	Reread the relevant Day 1-6 notes, redo worked examples, then sit a short retest.

RETURN FOR REVIEW

Send the completed paper and this marked score grid. The next lesson should be chosen from the lowest topic percentage, not the total score alone.